

Chapter 4

Preliminary Results

The preliminary results use the lowest fidelity model, the 0D scaled model, to generate improved DNP group parameters. Specifically, thermal fission of ^{235}U is used, since there is sufficient data in the literature to verify that the stationary case with no chemical removal produces expected results. These preliminary results investigate individual DNPs, the sensitivity of the DNP group parameters to different variables, and verification of the DNP group parameters.

Of the MSR physical phenomena considered, only the chemical removal has an impact on the 0D scaled model. The in-core and ex-core residence times do not change the delayed neutron yield per fission because the number of fission events that lead to that count rate also varies. Due to the simplification of the 0D scaled model, the source of each DNP scales at the same rate as the number of fissions linearly. The proposed 0D flow model will not have this linear response, as it will account for time-dependent decays and parasitic absorptions.

Specifically, for the scaled flux method applied to the cumulative fission yield, the scaling term for both is the same. Although this approach is rapid and enables more simulations to be executed, it also does not allow for the impact of circulation of the fuel salt to be incorporated into the improved DNP group parameter generation. For the MSRE, approximate measured residence times for the in-core and ex-core regions are 9 seconds and 16 seconds, respectively [71].

An additional consideration for these preliminary results is that pulse irradiation results are not included. Whether the results from the pulse irradiation simulation are combined via simultaneous solution of a larger matrix or separately solved and then combined, they improve the DNP group parameters for the shorter-lived groups [89]. This means that the longer-lived groups can be expected to be more accurate than the shorter-lived groups for the 0D scaled model, as it does not include pulse irradiation results. This is discussed further in Section 5.1.1.

These results, unless otherwise specified, use the parameters given in Table 4.1. The sampled simulations row indicates the number of non-linear least squares solves that generate DNP group parameters. For each sampled simulation, the emission probability, half-lives, and fission yields are sampled within their uncertainties using a normal distribution.

Table 4.1: Default parameters for 0D scaled model simulations.

Parameter	Value
Fissile Nuclide	^{235}U [-]
Incident Neutron Energy	0.0253 [eV]
Fission Yields	ENDF/B-VII.1 [34]
Half-lives	IAEA [50]
Emission Probability	IAEA [50]
Final Decay Time	600 [s]
Decay Time Spacing	Logarithmic
Decay Time Nodes	300
Sampled Simulations	5000

4.1 Important DNPs

Because the microscopic approach uses individual DNPs, it becomes possible to analyze the contribution of each DNP to the total delayed neutron yield. This is given in Equation (2.28), recreated here as:

$$\bar{v}_d = \sum_{i=1}^I P_{n,i} CFY_i.$$

Another quantity of interest is the number of delayed neutrons from the existing concentration of nuclides, rather than the number of delayed neutrons per fission. This can be calculated by integrating the delayed neutron count rate, as shown in Equation (2.27), recreated here as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0).$$

Figure 4.1 shows both of these approaches and how they differ. Figure 4.2(a) shows the delayed neutron yield contributions, which indicate the DNPs which emit the most delayed neutrons after a single fission event. Figure 4.2(b) shows the contribution of each DNPs to the delayed neutron count rate for a saturation irradiation. This does not show the delayed neutron yield, but contributes directly to the DNP group yields, as the delayed neutron count rate is critical for calculating the DNP group parameters. Figure 4.2(c) shows the concentrations of each DNP, which is correlated to Figure 4.2(b). DNPs with a large concentration may have a small emission probability and not contribute many delayed neutrons to the count rate, such as ^{141}Cs .

For the delayed neutron yield contributions, shown in Figure 4.2(a), ^{86}Ge has the largest contribution to the delayed neutron yield, which means that for each thermal fission event in ^{235}U , approximately 12.5% of the delayed neutrons come from ^{86}Ge . ^{86}Ge is very short lived, with a half-life of approximately 200 milliseconds. This is why, in Figure 4.2(b), it contributes less than 4% of the total delayed neutrons once the sample has been irradiated for some time. Since it has been irradiated, the longer lived nuclides have accumulated, and the delayed neutron count rate is dominated by them. This reinforces the justification of why pulse irradiation data is important, as the saturation irradiation does not fully capture the relative effects of the short-lived DNPs. Finally, the relative concentration of each DNP is shown in Figure 4.2(c). The concentrations are generally consistent with the literature, though ^{141}Cs is an outlier [99]. This may be due to the simplifications made in the 0D scaled model, such as exclusion of parasitic absorption, or that other works ignore ^{141}Cs due to its exceedingly small emission probability.

Figure 4.2 shows the delayed neutron count rate in Figure 4.2(a), as well as the integrated form in Figures 4.2(b) and 4.2(c) which represents the total delayed neutron counts at each time. Figure 4.2(a) shows how the count rate of ^{86}Ge dominates at short time scales due to its short half-life, while after a tenth of a second it becomes less

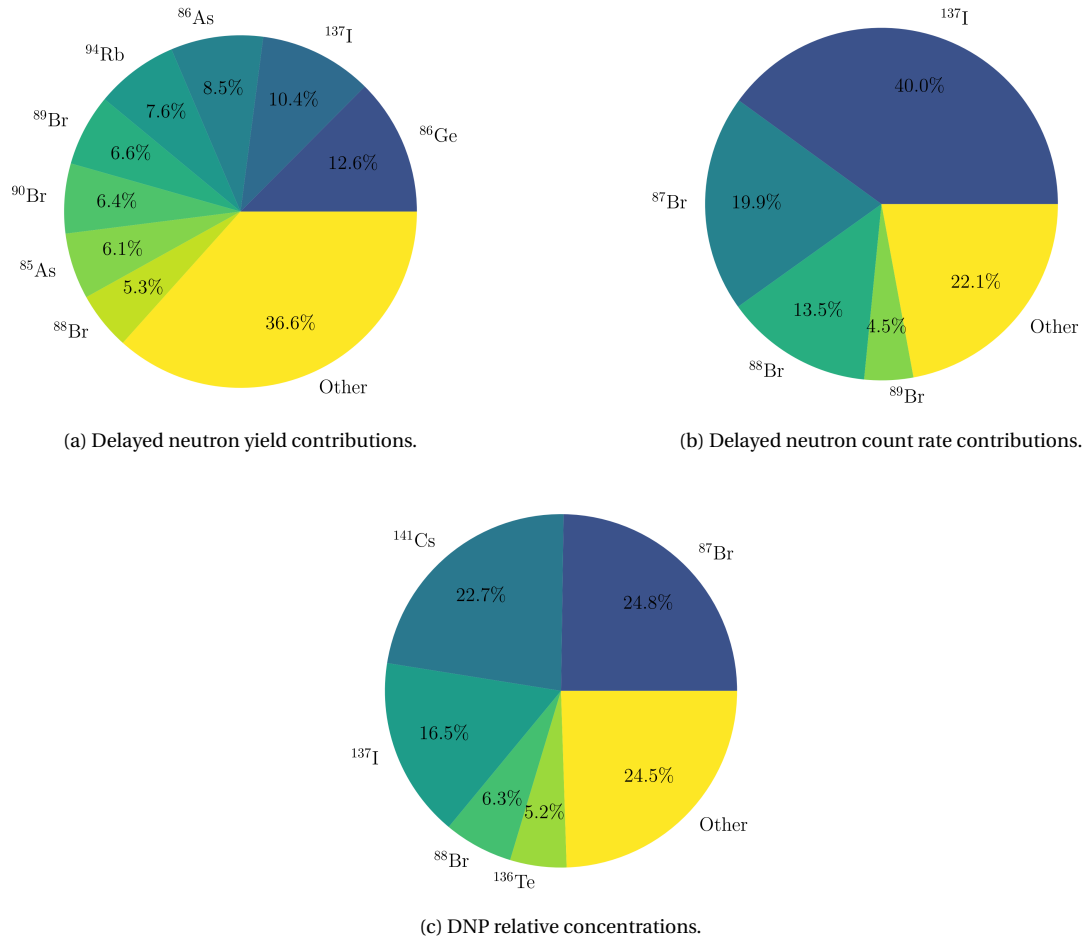
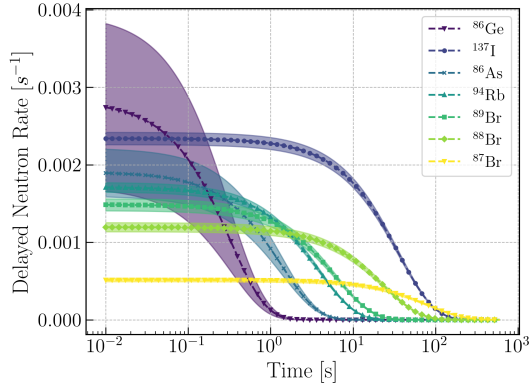


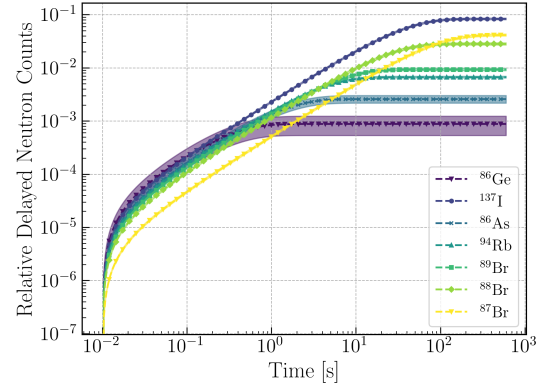
Figure 4.1: Delayed neutron contributions from each DNP after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

important. This again demonstrates how the saturation irradiation enables better analysis of the longer-lived DNPs, but at the cost of less relative importance of the shorter-lived DNPs. Figure 4.2(b) shows how the total number of counts vary over time on a log scale. The initial contribution of ^{86}Ge is initially dominant but quickly becomes relatively unimportant compared to the other DNPs. Figure 4.2(c) reflects Figure 4.1(b), as both figures show that the contribution of ^{86}Ge is minimal following a saturation irradiation. The total delayed neutron count contribution is dominated by longer lived DNPs like ^{137}I and ^{87}Br .

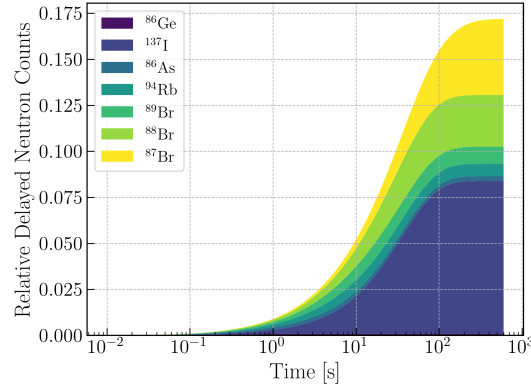
Another analysis of interest with the individual DNP data is calculation of the delayed neutron yield and average half-life, shown in Table 4.2. These values are calculated and then compared with the six group DNP group parameters formulated from those DNPs. Ideally these values should be identical, but because the six-group fit cannot fully capture the behavior of all the decaying DNPs, there are some small differences. These differences are small relative to the uncertainty bounds for each term.



(a) The delayed neutron count rate.



(b) The total delayed neutron counts.



(c) A stack plot of the total delayed neutron counts.

Figure 4.2: Delayed neutron count rate contributions from the top two most contributing DNPs at each time step after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Table 4.2: The delayed neutron yield and average half-life calculated by summing individual DNPs and from the 6 DNP group parameters.

Term	DNP	Group
$\bar{\nu}_d$	0.0225 ± 0.0015	0.0224 ± 0.0013
$\bar{\tau}[\text{s}]$	6.3852 ± 0.3801	6.4029 ± 0.3365

4.2 Sensitivity Studies

This section covers the sensitivity of the DNP group parameters to various simulation parameters. The simulation parameters evaluated are the chemical removal, the time nodes at which to evaluate the non-linear least squares problem, and the uncertainties in the nuclear data. For the chemical removal analysis, the MSBR chemical reprocessing scheme is used. The MSBR reprocessing scheme has large chemical removal rates, providing a conservative impact of these chemical removal rates on the DNP group parameters. The evaluation of the time nodes include the spacing between each node, the number of nodes, the total amount of time the sample should decay, and a combination of the total time and number of nodes. Finally, the data uncertainties include uncertainty in the fission yields, half-lives, and emission probabilities. To propagate these uncertainties to the DNP group

parameters, the non-linear least squares solver is run 5000 times with a random normal distribution sample taken from the fission yield, half-life, and emission probability of each DNP.

4.2.1 Chemical Removal

As shown in Appendix B, the fraction of fuel salt in the ex-core region determines the scaling of the chemical removal rate. For this analysis, the ex-core fraction will be assumed to be 0.5. This value is based on the approximate core zone I, core zone II, and annulus salt volume relative to the primary system salt volume in the MSBR [148]. The chemical reprocessing scheme from the MSBR used in this analysis is given in Table 3.1, recreated here.

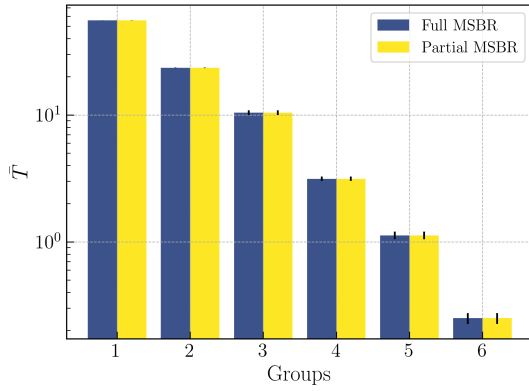
Table 4.3: MSBR Online Reprocessing Removal Rates from [148].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

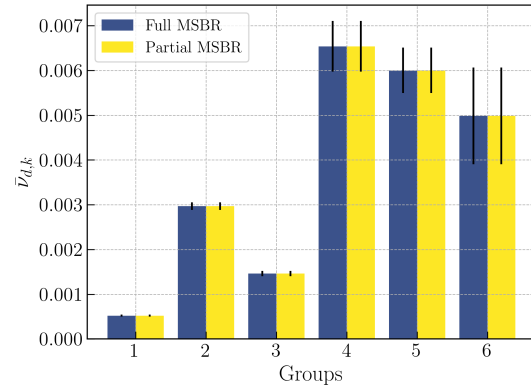
One consideration is that the 0D scaled model will underestimate the effect of the chemical removal rate. This is because it only chemically removes from the cumulative fission yield of the element in the reprocessing scheme, and does not account for decay chains which may have chemical removal rates included. For example, ^{137}I has a cycle time of 60 days, or a chemical removal rate of $1.9 \times 10^{-7} [s^{-1}]$. However, Sb and Te, which both decay into I, also have a chemical removal rate of $0.05 [s^{-1}]$, many orders of magnitudes larger than the I removal rate. However, using the 0D scaled model, with cumulative fission yields and no decay chain, means that this effect is not captured.

Long Cycle Time Evaluation

Based on the chemical removal rates provided in Table 3.1, there are many elements to consider. However, since the sample is only irradiated over the course of several minutes, it is of interest to determine the significance of the smaller removal rates, such as the 500 day cycle time of rare earths. For this analysis, only 20 second cycle time elements, the noble metals and gases, are considered. The difference between the full MSBR reprocessing scheme and the partial MSBR reprocessing scheme, with only 20 second cycle time elements considered, is shown in Figure 4.3. The difference between each DNP group parameter, $\bar{v}_{d,k}$ and λ_k , is less than 0.001%. This analysis



(a) Half-life DNP group parameters.



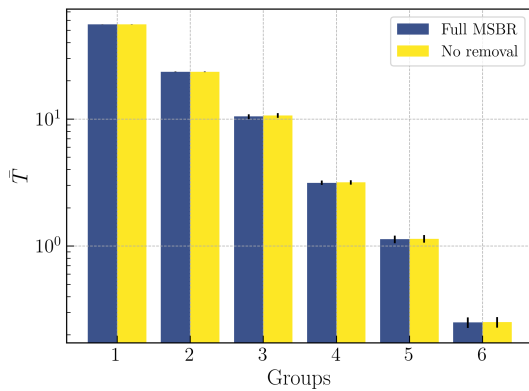
(b) Yield DNP group parameters.

Figure 4.3: Delayed neutron contributions from each DNP with and without long cycle time chemical removal rates in the MSBR after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

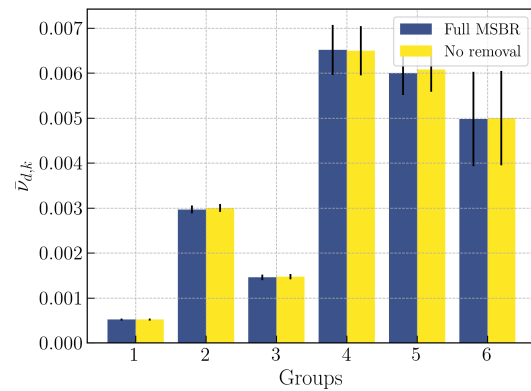
shows that the elements with cycle times longer than 20 seconds do not affect the DNP group parameters for a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Removal Rate Evaluation

It is also of interest to determine the sensitivity of the DNP group parameters to the chemical reprocessing scheme. The 0D scaled model will underestimate the effect of chemical removal due to the use of cumulative fission yields without explicit nuclide tracking, but this analysis can still give a useful conceptual understanding of the chemical reprocessing impact. The comparison of DNP group parameters with and without the MSBR reprocessing scheme is given in Figure 4.4. This figure shows that the half-lives are very similar, while the yields without removal are slightly higher.



(a) Half-life DNP group parameters.



(b) Yield DNP group parameters.

Figure 4.4: Delayed neutron contributions from each DNP with and without the MSBR chemical removal rates after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The difference between the average half-lives is negligible, while the difference between the total delayed

neutron yields is 0.57%, or approximately 12 *pcm*. This is a small difference, which is likely due to the nature of the 0D scaled model not treating the decay chain explicitly. However, since the effects of chemical reprocessing will consistently reduce the total delayed neutron yield due to removal of DNPs, it is important to account for this factor.

4.2.2 Time Nodes

The time node sensitivity is important, as many time nodes may be required to capture the physics fully, but at an increasing cost for each additional time node. Of interest are the distribution of nodes, the number of nodes, the amount of time the sample decays, and the combined decay time and number of nodes.

Distribution

For the distribution of time nodes, two different approaches are compared: a logarithmic spacing between 0.01 and 600 seconds, and a linear spacing between 0 and 600 seconds. Because 300 time nodes are being used, this means that the linear spacing will not capture behavior which occurs in the first 2 seconds, but instead will have a higher resolution at the longer time scales. This may not be as useful, as at longer time scales the number of DNPs that are relevant drop away quickly, as discussed in Section 4.1.

Figure 4.5 shows how the DNP group parameters compare between the logarithmic and linear spacing. The half-lives of the shorter lived groups are measured as being larger for the linear spacing, which is consistent with expectations. The linear spacing does not have any data between zero and two seconds, and there is greater weighting on the longer lived groups due to the data distribution. This also leads to the shorter lived groups capturing a larger yield, with the yields being pulled from groups two, three, and four to be added to groups five and six. The difference between the total DNP yields is approximately 0.1%, so changing the spacing of the time nodes does not change the macroscopic behavior of the DNP groups. Instead, it affects the distribution of delayed neutrons over time, where the logarithmic distribution is more accurate due to capturing the effects of shorter lived DNPs.

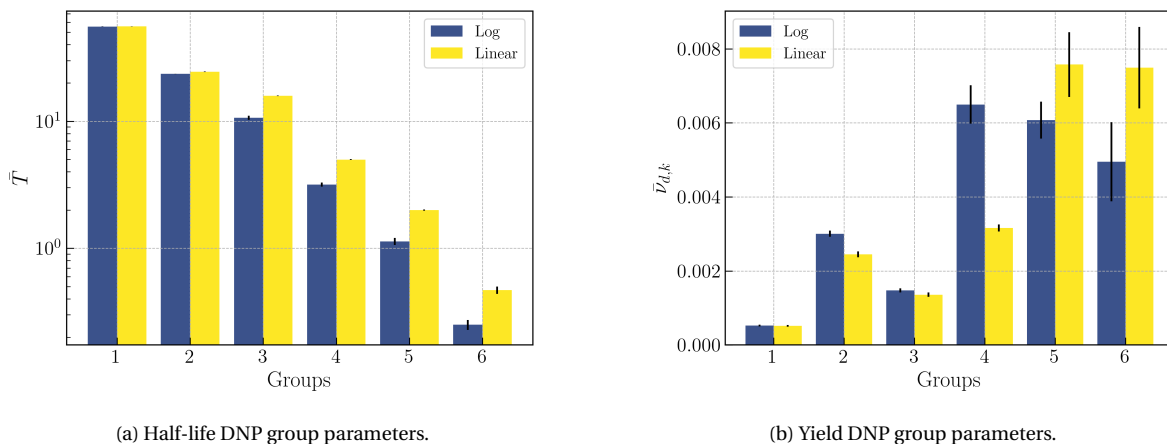


Figure 4.5: DNP group parameters for linear and logarithmic spacings of the time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Number of Nodes

Figure 4.6 shows how varying the number of nodes affects the results. It appears that there is no significant difference between the results when varying the number of nodes. The main difference appears in the yields of groups four and five, with increasing nodes increasing the yield of group four and decreasing the yield of group five. There are also some smaller differences, such as the yield of group six decreasing and the yields of groups two and three increasing with an increasing number of nodes.

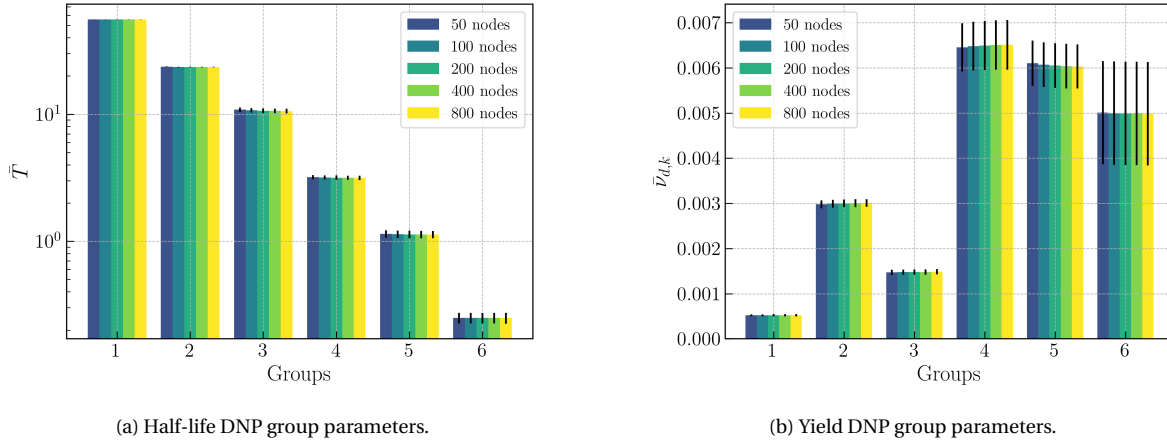


Figure 4.6: DNP group parameters for varying numbers of time nodes after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

The simulation with the most nodes is the most accurate, so the percent difference of the DNP group yields for each group can be calculated, as well as the computational cost associated with the increase in accuracy. This is shown in Table 4.4, where using 50 nodes results in the group five yield being off by approximately 1%, and using 200 nodes keeps differences below 0.02%. The computational cost is also fairly constant for each of these different numbers of nodes. Based on these results, the selected default of 300 nodes appears to offer sufficiently high accuracy yields, within 0.1%, while also residing in the constant time cost regime.

Table 4.4: Comparison of how the number of nodes changes the computational cost and relative difference in yields compared to the 400 node case after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Nodes	Yield Difference [%]						Cost [s]
	1	2	3	4	5	6	
50	0.30	0.90	0.53	0.75	0.99	0.59	297
100	0.17	0.49	0.25	0.39	0.46	0.77	297
200	0.01	0.02	0.00	0.01	0.01	0.09	300
400	0.00	0.00	0.00	0.00	0.00	0.01	304
800	-	-	-	-	-	-	310

Total Decay Time

As the total decay time increases, two effects occur at the same time: a decrease in the time node density, discussed in Section 4.2.2, and an increase in the number of nodes evaluated at longer times. Because the longest lived DNP, ^{87}Br , has a half-life of approximately one minute, it is reasonable to assume that the default final time of ten minutes

offers sufficient data on all DNP decays.

Figure 4.7 shows that this is true. The differences in DNP group parameters between a 600 second decay time and a 1000 second decay time are very small relative to the uncertainty bounds. The short decay times do not accurately model the DNPs, with underestimations of the half-lives. The half-life underestimations are then correlated with overestimation of some of the longer lived DNP group yields. The DNP yield of groups two and four have an interesting trend, where they underestimate for short decay times and overestimate for intermediate decay times. This is likely due to the simultaneous fitting of the surrounding groups, as the raw data itself remains unchanged.

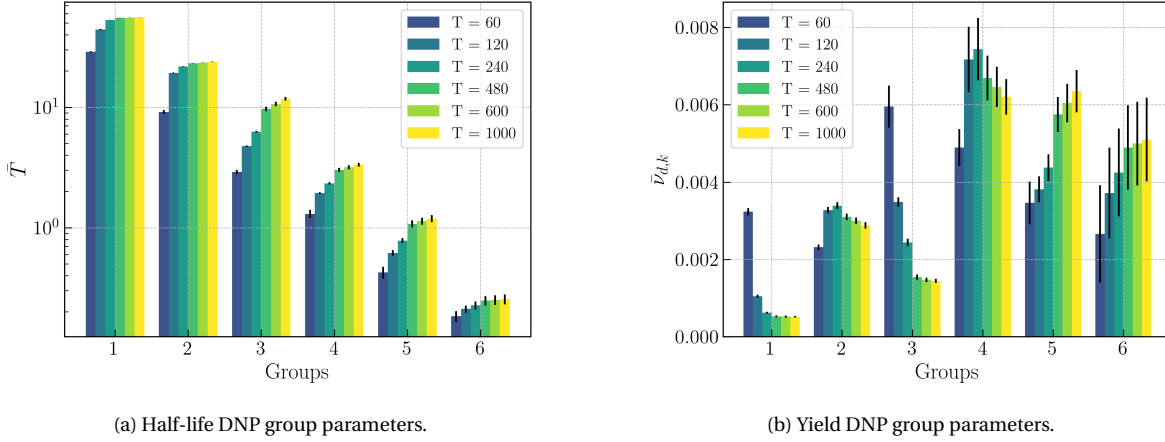


Figure 4.7: DNP group parameters for varying total decay times after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

Unfortunately, it cannot be claimed that increasing the net decay time strictly improves the accuracy of the DNP group parameters. Excessively short decay times would not properly capture long-lived DNPs, while excessively long decay times would overfit the long-lived DNPs. Another way to compare these values is using the total delayed neutron yields and average half-lives, shown in Table 4.5. From this table, it can be seen that the differences in these values are small and within uncertainty bounds of each other.

Table 4.5: Comparison of how the net decay time, T changes the total delayed neutron yield and average half-life after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

T [s]	$\bar{v}_d$	$\bar{\tau}$ [s]
60	0.0227 ± 0.0016	6.1926 ± 0.4058
120	0.0227 ± 0.0015	6.3308 ± 0.3804
240	0.0227 ± 0.0015	6.3581 ± 0.3651
480	0.0226 ± 0.0013	6.3647 ± 0.3432
600	0.0226 ± 0.0013	6.3648 ± 0.3424
1000	0.0226 ± 0.0013	6.3647 ± 0.3426

Time Node Density

Section 4.2.2 shows how varying the total decay time affects the results, and Section 4.2.2 shows how varying the number of time nodes affects the results. This section evaluates the sensitivity of the DNP group parameters to the node density, number of nodes, and decay time by comparing varying numbers of nodes and decay times. Figure 4.8 shows how the DNP group parameters vary for varying node densities.

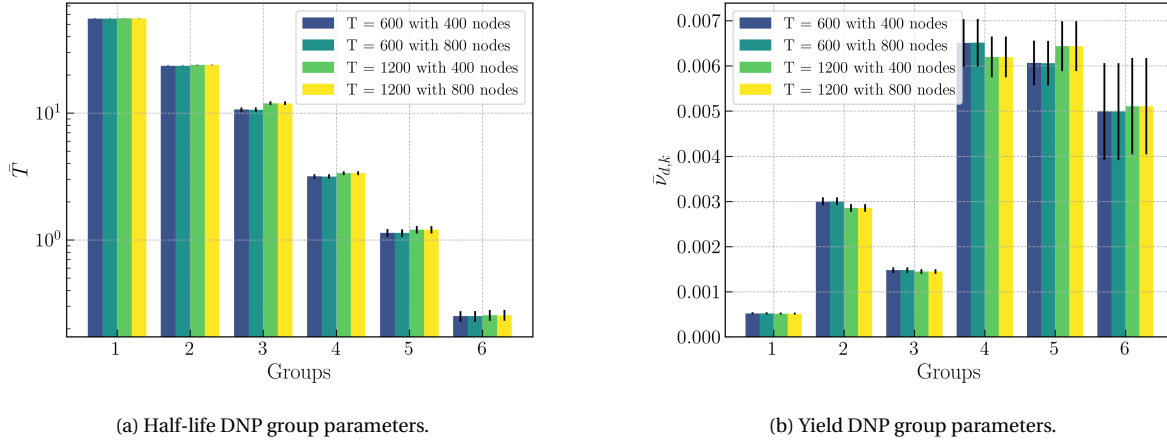


Figure 4.8: DNP group parameters for varying node densities after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

If the node density was the driving factor in the DNP group parameters, then the decay time of 600 seconds with 400 time nodes would be equivalent to the decay time twice as long and with twice as many time nodes. Instead, the results are approximately constant with varying time nodes, and the main difference is in the final decay time. This indicates that the sensitivity analysis discussed in Section 4.2.2 is valid, and does not need to also account for the varying time node density.

This raises the same issue as in Section 4.2.2, as increasing the total decay time does not appear to necessarily make the results more accurate. This can be seen as the group 2 through 6 DNP half-lives strictly increase as the total decay time increases due to the increased number of data points at longer times. The proportional distribution of nodes remains unchanged between shorter-lived and longer-lived times, but it is currently unclear what decay time should be selected. Further analyses with the 0D flow model should make this analysis more easily verifiable with other results in the literature.

4.2.3 Nuclear Data Uncertainties

To find the sensitivity of the DNP group parameters to the uncertainties in the nuclear data, a Monte Carlo approach is used. From each piece of data used in the 0D scaled model, such as the half-life of ^{87}Br , a random value is sampled assuming a normal distribution. This is repeated for every DNPs' fission yield, half-life, and emission probability.

Count Rate

The count rate, normalized to the Keepin et al. count rate, is given in Figure 4.9. The red circles indicate the count rates from various sampled simulations, while the blue dashed line indicates the DNP group parameters used and the uncertainty based on the 1000 Monte Carlo sampled non-linear least squares solves. The black X's indicate the mean count rate at each point. The other six group DNP parameters plotted include Keepin, which is 1.0 since the plot is normalized to the Keepin values, the Brady group parameters, and the Synetos group parameters [89, 23, 179].

This figure shows that beyond roughly three seconds, there is agreement within the uncertainty bounds between the current DNP group parameters and most of the other results in the literature, with the Brady group parameters having a noticeable difference at longer times. For the short time scales, as previously discussed in the introduction

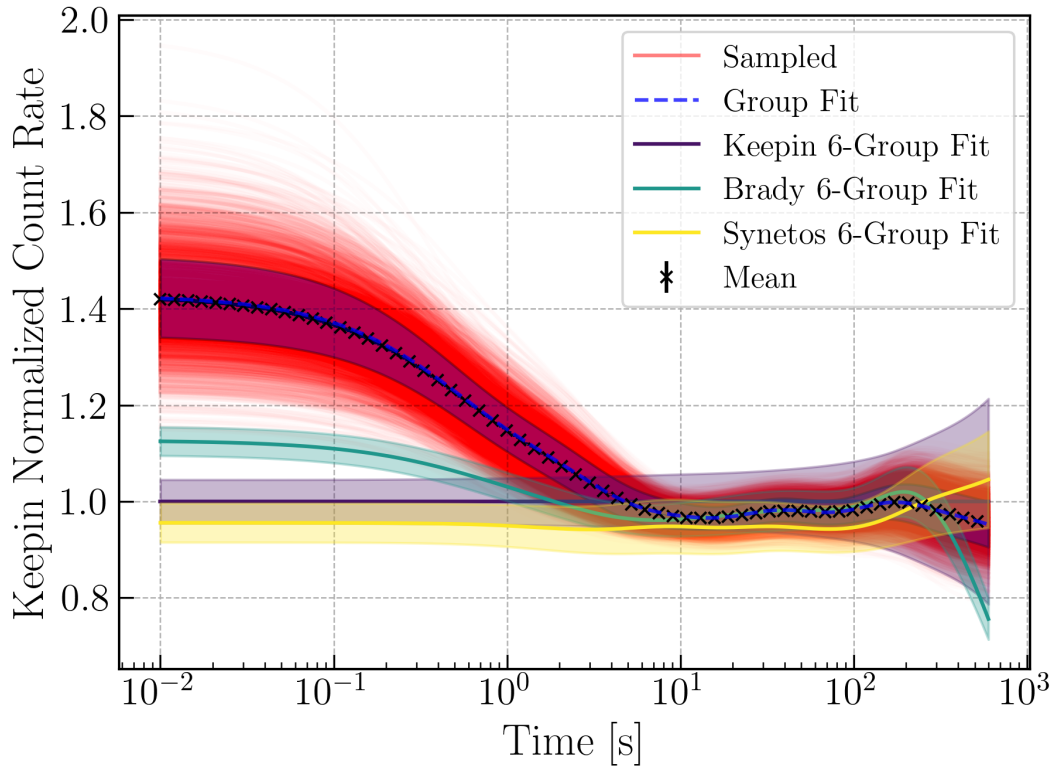


Figure 4.9: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

to this chapter, are inaccurate due to the lack of pulse irradiation data. It is expected that once this data is added, the earlier times will fit more closely. To demonstrate this, the first two groups are used while the shorter-lived groups three through six are replaced with the Keepin et al. DNP group parameters [89]. This modified set of DNP group parameters can be seen in Figure 4.10.

The modified DNP group parameters are not a perfect comparison, as the non-linear least squares solve adjusts the yields of each group relative to the others and based on the half-lives of the others. This means that not including pulse irradiation data will also affect the longer-lived groups, though by a smaller amount than the effects on the shorter-lived groups. However, there is generally better agreement when adjusting the shorter lived groups, particularly during the times less than one second.

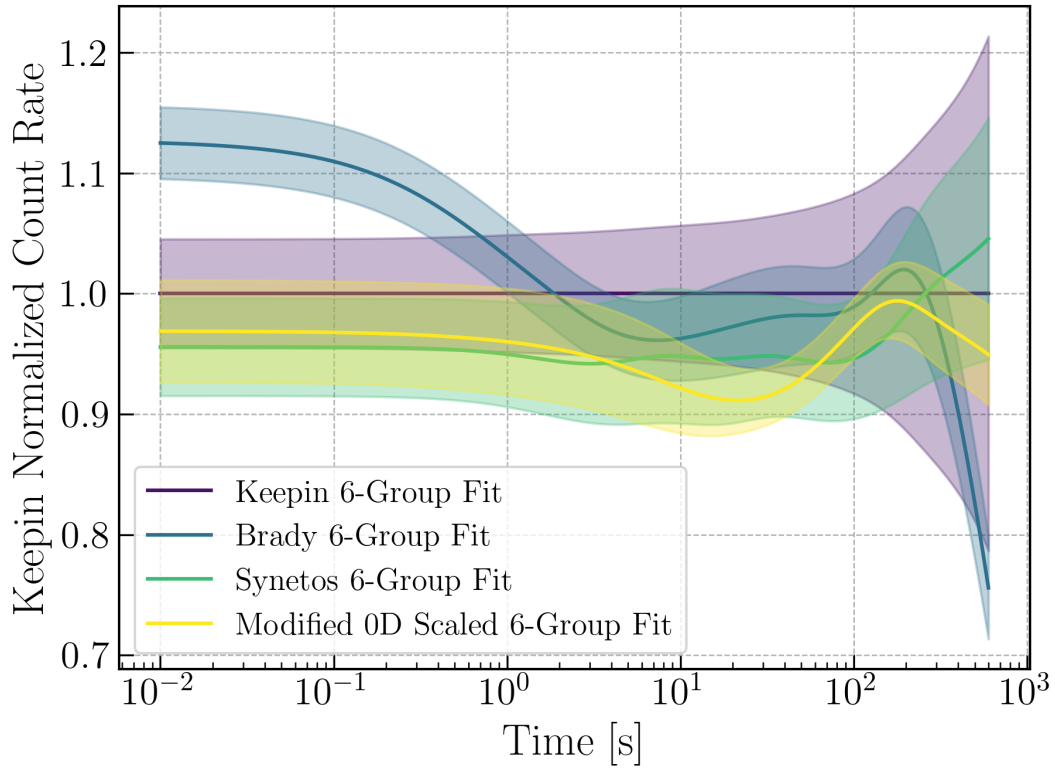


Figure 4.10: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

Individual DNPs

The Monte Carlo sampling of each DNPs' nuclear data allows for sensitivity analyses on how each change affects the DNP group parameters. For example, since group 1 is dominated by ^{87}Br , then changes to the half-life, emission probability, or fission yield of ^{87}Br would lead to noticeable changes in the group 1 DNP half-life and yield.

The number of Monte Carlo simulations is increased to 5000 for this analysis, as the contribution of each DNP needs sufficient data to ensure differences in DNP group parameters are caused by its own sampled variation. To find the nuclides with the most impactful sensitivities, a linear regression fits the slope of the relative change in DNP group parameters to the relative change to the individual DNP nuclear data, while also calculating the Pearson correlation coefficient (PCC). The PCC describes the linear relationship between two variables on a scale from -1, representing a perfect negative correlation, +1 is a perfect positive correlation, and 0 is no correlation [62]. The changes in nuclear data include changes to emission probability, P_n ; fission yields, which affect the concentration, N ; and half-lives, τ .

Sensitivities of interest in this analysis are for absolute values of the PCC greater than 0.2. This value is selected somewhat arbitrarily, but gives a useful indication that a nuclide has a non-negligible correlation with the DNP group parameter of interest. The results of this analysis are given in Table 4.6.

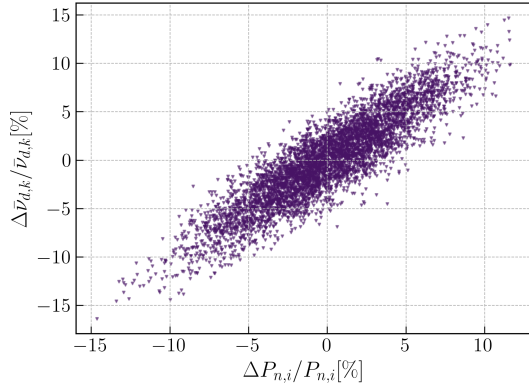
One of the most predictable sensitivities in Table 4.6 is the effect of ^{87}Br on the group one half-life and yield, as ^{87}Br dominates this group. Specifically, the sensitivities in ^{87}Br data that affect the DNP group parameters with a PCC greater than 0.2 are the emission probability on the group yield, the concentration on the group yield, and

Table 4.6: PCCs with a magnitude larger than 0.2 for variation in DNP data which correspond to changes in DNP group parameters after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

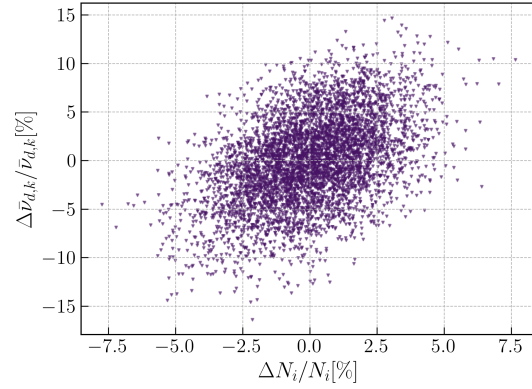
Nuclide	Group Value	DNP Value	PCC
^{84}Ga	$\tau_3[\text{s}]$	$N_i[-]$	-0.211724
^{84}Ga	$\tau_4[\text{s}]$	$N_i[-]$	-0.222001
^{84}Ga	$\tau_5[\text{s}]$	$N_i[-]$	-0.203050
^{85}As	$\bar{v}_{d,4}[-]$	$N_i[-]$	0.757084
^{85}As	$\tau_4[\text{s}]$	$N_i[-]$	-0.411166
^{85}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.718393
^{85}As	$\tau_5[\text{s}]$	$N_i[-]$	0.448911
^{88}Br	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.276294
^{88}Br	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.221999
^{88}Br	$\tau_3[\text{s}]$	$P_{n,i}[-]$	0.379647
^{88}Br	$\tau_3[\text{s}]$	$N_i[-]$	0.264940
^{138}I	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.211254
^{137}I	$\bar{v}_{d,2}[-]$	$P_{n,i}[-]$	0.478664
^{137}I	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.270646
^{137}I	$\bar{v}_{d,2}[-]$	$N_i[-]$	0.746893
^{137}I	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.813782
^{137}I	$\tau_2[\text{s}]$	$N_i[-]$	0.252540
^{86}As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.512737
^{98}Y	$\bar{v}_{d,4}[-]$	$P_{n,i}[-]$	0.243371
^{86}Ge	$\bar{v}_{d,2}[-]$	$\tau_i[\text{s}]$	-0.238309
^{86}Ge	$\tau_2[\text{s}]$	$\tau_i[\text{s}]$	0.359997
^{86}Ge	$\bar{v}_{d,3}[-]$	$\tau_i[\text{s}]$	-0.637401
^{86}Ge	$\tau_3[\text{s}]$	$\tau_i[\text{s}]$	0.656167
^{86}Ge	$\bar{v}_{d,4}[-]$	$\tau_i[\text{s}]$	-0.407397
^{86}Ge	$\tau_4[\text{s}]$	$\tau_i[\text{s}]$	0.673363
^{86}Ge	$\bar{v}_{d,5}[-]$	$\tau_i[\text{s}]$	0.263800
^{86}Ge	$\tau_5[\text{s}]$	$\tau_i[\text{s}]$	0.662307
^{86}Ge	$\bar{v}_{d,6}[-]$	$P_{n,i}[-]$	0.830397
^{86}Ge	$\bar{v}_{d,6}[-]$	$N_i[-]$	0.464433
^{86}Ge	$\tau_6[\text{s}]$	$P_{n,i}[-]$	-0.306136
^{86}Ge	$\tau_6[\text{s}]$	$\tau_i[\text{s}]$	0.893433
^{87}Br	$\bar{v}_{d,1}[-]$	$P_{n,i}[-]$	0.896423
^{87}Br	$\bar{v}_{d,1}[-]$	$N_i[-]$	0.456307
^{87}Br	$\tau_1[\text{s}]$	$\tau_i[\text{s}]$	0.997442
^{137}Sb	$\bar{v}_{d,3}[-]$	$N_i[-]$	0.229445
^{137}Sb	$\tau_3[\text{s}]$	$N_i[-]$	-0.229578
^{137}Sb	$\tau_4[\text{s}]$	$N_i[-]$	-0.250857
^{137}Sb	$\tau_5[\text{s}]$	$N_i[-]$	-0.299691

the half-life on the group half-life. Figure 4.11 shows how the group one parameters vary with changes in the ^{87}Br data, as well as less strongly correlated data, such as the emission probability of ^{87}Br on the group three half-life. Although it is not intuitive that changes to the emission probability of ^{87}Br would affect the half-life of group three, there is a non-zero correlation. This is because the DNP group parameters cannot be taken in isolation, but as a collective set of parameters that best fit the delayed neutron count rate curve. Thus, changes to that curve propagate to all group parameters, albeit with varying magnitudes to different groups and different parameters.

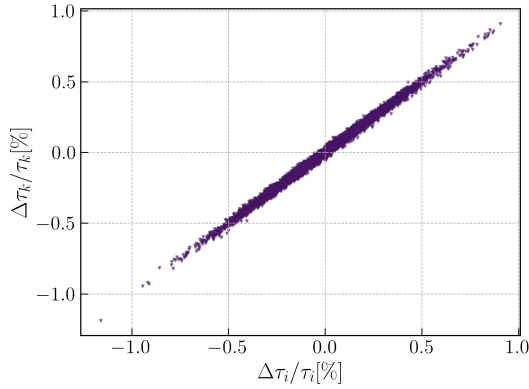
Another nuclide of interest in Table 4.6 is ^{86}Ge , as it is also an important nuclide for calculating delayed neutron yield contributions as shown in Figure 4.1. It can be seen that ^{86}Ge has an impact on many DNP group parameters



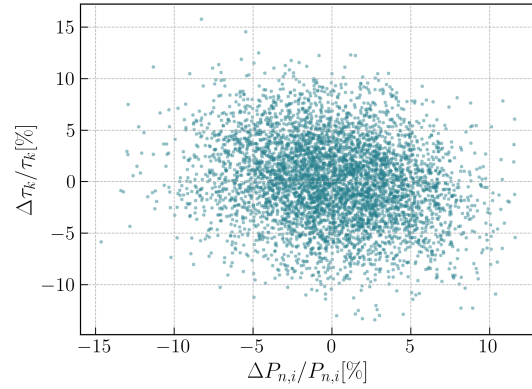
(a) DNP group 1 yield variation due to emission probability variation with a PCC of 0.90.



(b) DNP group 1 yield variation due to fission yield variation with a PCC of 0.46.



(c) DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.



(d) DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

Figure 4.11: DNP group parameter sensitivity to changes in ^{87}Br nuclear data after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

as there is variation in different parameters, especially its half-life. Figures 4.12 and 4.13 show the sensitivity to each DNP group parameter to each variation in data for ^{86}Ge .

Figure 4.12 has the largest sensitivities for the group half-lives when there are changes in the half-life of ^{86}Ge . It is interesting to note that, for increasingly longer-lived groups, the impact of the half-life of ^{86}Ge on the group half-life becomes less and less. This is intuitive, as a short-lived nuclide such as ^{86}Ge should have much more impact on short-lived groups than on long-lived groups. Another interesting pattern is that the group six half-life of ^{86}Ge sensitivity appears to have a line below which there are no data points, indicating a lack of symmetry. Sensitivities are typically symmetrical about the line that best fits the data for a linear response. This asymmetry is likely due to the fact that there are many longer-lived DNPs, whereas ^{86}Ge is one of the few very short-lived DNPs that dominates in this time region. Because there aren't shorter-lived DNPs in this time region, the half-life of group 6 does not decrease as much as it increases.

Figure 4.13 shows there are changes in the group yields based on changes in the half-life of ^{86}Ge . There is not a clear trend moving from shorter to longer-lived groups, though there are some negative correlations for groups three and four, which may indicate that a longer-lived ^{86}Ge leads to a decreased yield in those groups while also

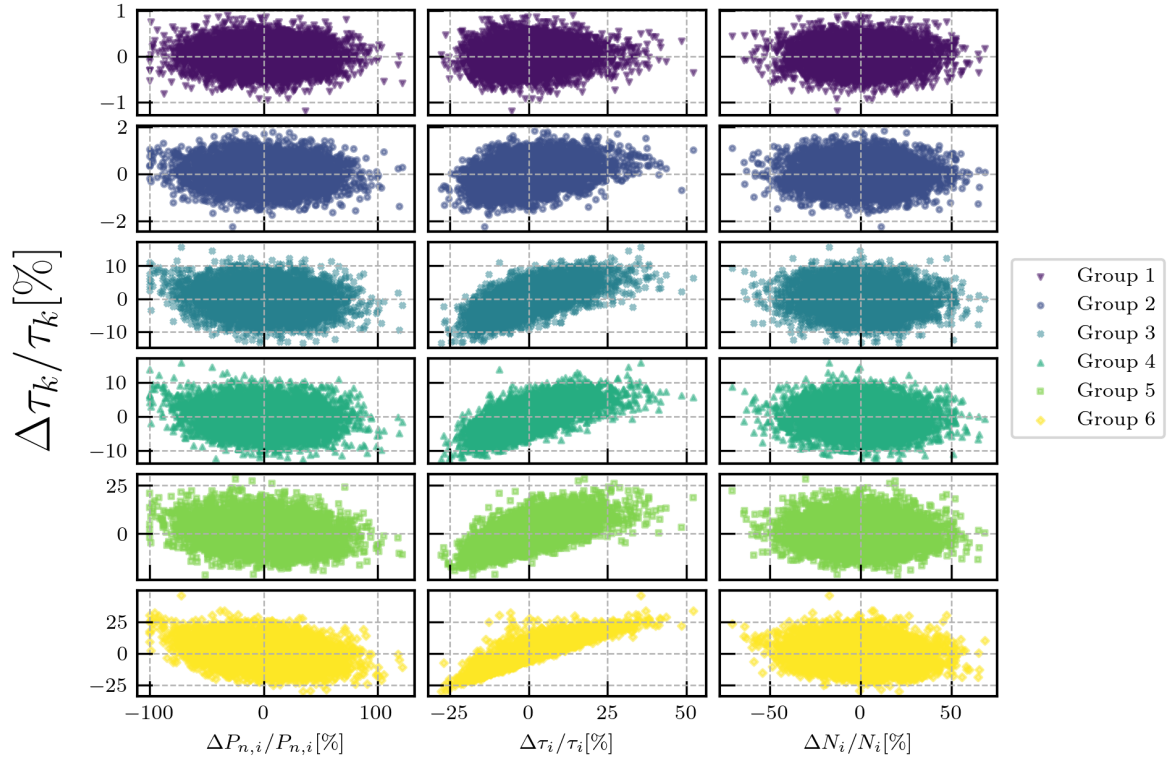


Figure 4.12: The difference in each DNP group half-life from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

increasing their half-lives, as shown in Figure 4.12. It appears those groups capture fewer short-lived DNPs as the half-life of ^{86}Ge increases both by increasing their group half-lives and decreasing their yields.

Another interesting sensitivity correlation is the emission probability of ^{86}Ge on the group six yield, which appears to have a narrow spread of values for small emission probabilities but then spreads out for larger emission probabilities. The trend is positive because increasing the emission probability for the short-lived ^{86}Ge corresponds to an increase in the short-lived group 6 yield. One contributing factor to this spreading out of the yield is likely the lack of pulse irradiation data. Because the short-lived groups are less accurate without this data, removing ^{86}Ge leads to a more consistent yield in group six. However, as ^{86}Ge becomes more impactful, the possible value for the yield spreads out due to the lack of pulse irradiation data.

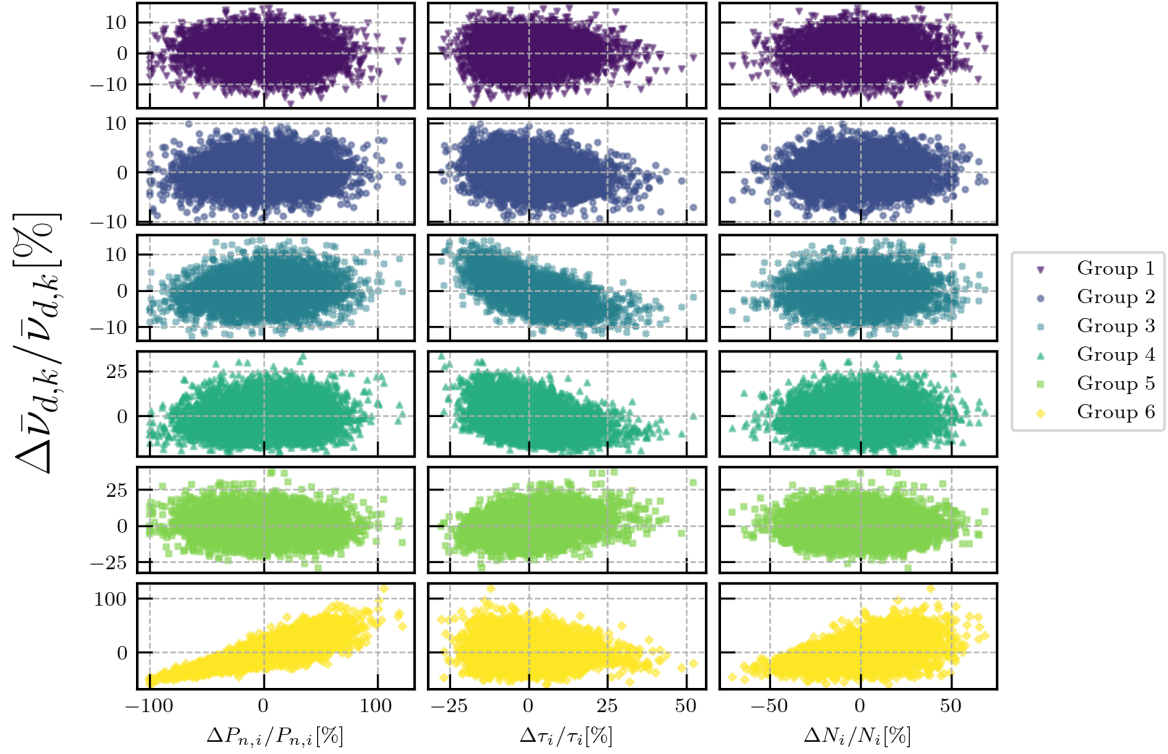


Figure 4.13: The difference in each DNP group yield from the ^{86}Ge relative changes in emission probability, half-life, and concentration after a thermal saturation irradiation of ^{235}U using the 0D scaled model.

4.3 Verification

The 0D scaled model is verified using DNP group parameters from thermal fission of ^{235}U . This verification uses the data from literature, introduced in Section 2.1.1, combined and recreated here with the results from this work included.

Tables 4.7, 4.8, 4.10, and 4.10 show the DNP group parameters from literature and how they compare to the 0D scaled model DNP group parameters. As previously discussed, because the 0D scaled model does not include the pulse irradiation data, the shorter-lived groups are expected to be less accurate. This can be seen in the yields, where the groups 5 and 6 yields are significantly different, while groups 1 and 2 are much more similar to results from the literature. The group half-lives are similar to the literature values, though group three is an outlier. This is again likely due to the pulse irradiation data missing, as it is noticeably longer than the other values in the literature.

Table 4.11 shows how the total delayed neutron yield for 100 fissions compares between the literature and for the 0D scaled model. The differences between the microscopic and macroscopic delayed yields for the same dataset exist for many fissile nuclides, not only ^{235}U [113]. The 0D scaled model yield is noticeably larger than the other values in the literature, but the raw data is also larger. This indicates not an issue with the DNP group parameter formulation, but rather, that the pulse irradiation data is necessary to capture the smaller yields of the shorter-lived DNP groups. This can be seen in Table 4.7, where the group 5 and 6 yields are greatly over-valued relative to other DNP group yields in the literature. Using the modified 0D scaled model, where groups three through six use the

Table 4.7: DNP group yields from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [36]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
ENDF/B-VI.8 [153]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [119]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [23]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158
Saleh et al. [160]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [179]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [187]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[192]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070
0D scaled model	0.00052	0.00297	0.00146	0.00650	0.00597	0.00500

* Modified to include a sixth group by Parish et al. [128].

Table 4.8: DNP group yield uncertainties from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [36]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [23]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009
Saleh et al. [160]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [179]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [187]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[192]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009
0D scaled model	0.00002	0.00009	0.00006	0.00056	0.00051	0.00111

* Modified to include a sixth group by Parish et al. [128].

Table 4.9: DNP group half-lives from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [36]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
ENDF/B-VI.8 [153]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [119]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [23]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462
Saleh et al. [160]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [179]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [187]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[192]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
0D scaled model	55.52977	23.54278	10.43942	3.14523	1.12586	0.25015

* Modified to include a sixth group by Parish et al. [128].

Keepin et al. DNP group parameters, results in a yield which is more consistent with the other values in the literature.

Table 4.10: DNP group half-life uncertainties from thermal fission of ^{235}U (recreated from [128]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [89]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [36]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
ENDF/B-VI.8 [153]	-	-	-	-	-	-
Mills et al. [119]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [23]	-	-	-	-	-	-
Saleh et al. [160]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [179]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [187]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[192]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
0D scaled model	0.15676	0.12560	0.44990	0.12771	0.07629	0.02325

* Modified to include a sixth group by Parish et al. [128].

Table 4.11: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [81]).

Approach	Data	Delayed Yield
Microscopic	JEF-2.2 [11]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [161]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [34]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [153]	1.670
Microscopic	Mills et al. [119]	1.645 ± 0.07
Microscopic	Brady and England [23]	1.780 ± 0.05
Microscopic	0D Scaled Model Data	2.254 ± 0.15
Microscopic	0D Scaled Model	2.256 ± 0.13
Hybrid	Modified 0D Scaled Model	1.531 ± 0.09
Macroscopic	JEF-2.2 [11]	1.655
Macroscopic	JEF-3.1.1 [161]	1.620
Macroscopic	ENDF/B-VII.0 [33]	1.585
Macroscopic	Keepin et al. [89]	1.580 ± 0.07
Macroscopic	Charlton et al. [36]	1.577 ± 0.03
Macroscopic	Saleh et al. [160]	1.645 ± 0.11
Macroscopic	Synetos and Williams [179]	1.510 ± 0.06
Macroscopic	Tuttle [187]	1.700 ± 0.05
Macroscopic	Waldo et al.[192]	1.670 ± 0.07

Table 4.12 shows how the average half-lives compare between the 0D scaled model and the literature. The 0D scaled model average half-life is too short, which is primarily due to the overvaluing of the group 5 and 6 yields, which weighs those half-lives more than they should be. Using the modified 0D scaled model with groups three through six replaced by the Keepin et al. DNP group parameters once again results in the model aligning much more closely with the literature. This indicates, once again, the importance of the pulse irradiation data to properly capture the short-lived DNP behavior.

Table 4.12: Average half-life from thermal fission of ^{235}U .

Approach	Data	Average half-life [s]
Microscopic	ENDF/B-VI.8 [153]	7.66
Microscopic	Mills et al. [119]	8.63 ± 0.52
Microscopic	Brady and England [23]	7.90 ± 0.34
Microscopic	0D Scaled Model Data	6.39 ± 0.38
Microscopic	0D Scaled Model	6.39 ± 0.34
Hybrid	Modified 0D Scaled Model	8.74 ± 0.46
Macroscopic	Keepin et al. [89]	9.03 ± 0.63
Macroscopic	Charlton et al. [36]	8.92 ± 0.28
Macroscopic	Saleh et al. [160]	9.49 ± 0.90
Macroscopic	Synetos and Williams [179]	8.96 ± 0.56
Macroscopic	Tuttle [187]	8.84 ± 0.45
Macroscopic	Waldo et al.[192]	8.72 ± 0.73